



PYTHON

USING APIS WITH REQUESTS

WHAT IS AN API? TALKING TO THE INTERNET WITH PYTHON



HAVE YOU USED ZOMATO OR SWIGGY? PYTHON TALKS TO THE INTERNET, TOO!

Think about when you order food at home. You open an app, pick your favorite biryani or pizza, and tap "Order"! But how does the restaurant know what you want? There's something special happening behind the scenes — and Python can do the same thing!

Python can talk to internet apps too. It sends messages and gets answers back — like having a super-smart helper that can ask questions anywhere online!



**PYTHON CAN TALK TO APPS
JUST LIKE YOU DO!**

LET'S EXPLORE!

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WHAT IS **AN API?**

THE RESTAURANT WAITER ANALOGY

An API is like a helpful waiter! When you want something from a restaurant kitchen, you don't go inside yourself. You tell the waiter what you want, and the waiter brings it back. Python uses APIs the same way to get information from the internet!

API = WAITER IN A RESTAURANT!

You tell the waiter your order. The waiter takes your order to the kitchen. The kitchen makes the food. The waiter brings food back to you. That's exactly how APIs work!

PYTHON IS YOU, API IS THE WAITER!



REQUEST AND RESPONSE — HOW APIS WORK



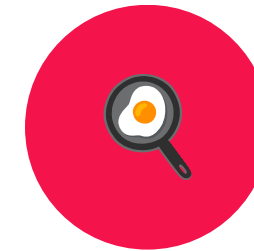
YOU = PYTHON PROGRAM

You decide what food you want to order. Your Python code decides what data to ask for!



WAITER = API

The waiter takes your order to the kitchen. The API carries your request to the server!



KITCHEN → FOOD = SERVER → RESPONSE

Kitchen makes food and sends it back. Server processes the request and sends data back!

Think of it like ordering chai: You say "Brother, one cutting chai!" — that's your REQUEST. The tea seller gives you tea — that's the RESPONSE. Status Code 200 = Success, tea ready! If there's no milk today, that's an ERROR RESPONSE — Status Code 404 = Not Found. The URL is the tea seller's address where you go to order.



JSON DATA

FORMAT

JSON is like your Aadhaar card data! It stores information neatly:

- Name
- Date of birth
- City
- ID number

JSON is what computers share with each other!

DATA ORGANIZED LIKE AN AADHAAR CARD!

```
{  
  'name': 'Priya Sharma',  
  'dob': '2015-04-12',  
  'city': 'Bengaluru',  
  'aadhaar': '1234-5678-9012'  
}
```

It looks just like a Python dictionary!

COMPUTERS LOVE JSON!



REQUESTS LIBRARY

YOUR JIO SIM!

PYTHON NEEDS REQUESTS LIKE YOU NEED A JIO SIM TO MAKE CALLS!

The requests library is Python's special tool to talk to the internet. Just like your phone needs a SIM card to call your friends, Python needs the requests library to call websites and APIs!

INSTALL IT FIRST:

```
pip install requests
```

IMPORT AT TOP:

```
import requests
```

ASK FOR DATA:

```
requests.get(url)
```

LET'S LEARN!



LIVE CODING

LET'S WRITE SOME CODE!

STEP 1: IMPORT REQUESTS

Get ready for some fun coding! We're going to write a Python program that asks the internet for a joke — yes, really! We start by importing the requests library. This is always the first line in our code when we want Python to talk to the internet.

TYPE THIS LINE AT THE TOP OF YOUR CODE!

```
import requests
```

READY? LET'S GO!



DEFINE THE URL

ADDRESS

The URL is like a phone number for the internet! It tells Python exactly where to go to ask for our joke. Every API has its own special address — think of it like dialing a friend's number.

THE API URL

```
url = "https://official-joke-  
api.appspot.com/random_joke"
```

Python asks this address for a joke — just like dialing a friend!



SENDING THE REQUEST & READING THE RESPONSE

`requests.get()` sends our request to the API server. The 'response' is what the API sends back — like getting a reply to your message! We then turn that response into something Python can read.

GET THE JOKE:

```
response = requests.get(url)
data = response.json()
```

This sends the request and reads the joke into Python as a dictionary!

PRINT THE JOKE:

```
print(data['setup'])
print(data['punchline'])
```

'setup' shows the joke's question; 'punchline' shows the funny answer!



ACTIVITY TIME!

YOUR TURN!

TRY THE JOKE API

NOW IT'S YOUR TURN TO CODE!

You've learned how Python talks to APIs. Now try it yourself! Open your code editor, type the joke code we learned, and run it. Can you get a different joke each time? Try running it 3 times and see what happens!



YOU JUST MADE PYTHON TALK TO THE
INTERNET — AMAZING!

KEEP GOING!

01

02

03



SUMMARY — WHAT WE LEARNED!

API

A waiter who carries requests and responses between you and the server.

REQUEST / RESPONSE

Request = asking for data. Response = getting data back from the API.

JSON / REQUESTS

JSON = data format like an Aadhaar card.
requests = Python's API tool.

API VOCABULARY FLASHCARDS

API

Waiter between you and data

REQUEST

Asking for something

RESPONSE

Answer from the API

URL / ENDPOINT

The API's address, like a house address

STATUS CODE

200 = success, 404 = not found

JSON

Data format computers share, like a dictionary